# Predicting market volatility risk

## Can liquidity and market signals identify high-risk trading days?

ETF liquidity proxies · VIX · Machine learning

An individual applied risk analytics project

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### ANALYSIS FRAMEWORK

01 Integrate signals

02 Identify stress regimes

03 Predict next-day risk

04 Evaluate alerts

# Why an early-warning signal matters

## Problem

Large next-day price moves can affect a portfolio before analysts can react.

## Target

High risk = |next-day SPY return| above 1.5%.

## Decision

Send a monitoring alert rather than trigger an automatic trade.

**ILLUSTRATIVE COST ASSUMPTIONS**

**Missed high-risk day = 5 units • False alarm = 1 unit**

# Two information streams, one modeling panel

## 01 Market trading data

SPY + QQQ daily OHLCV

Amihud illiquidity

Corwin–Schultz spread proxy

Rolling return volatility

+

## 02 Market-risk signal

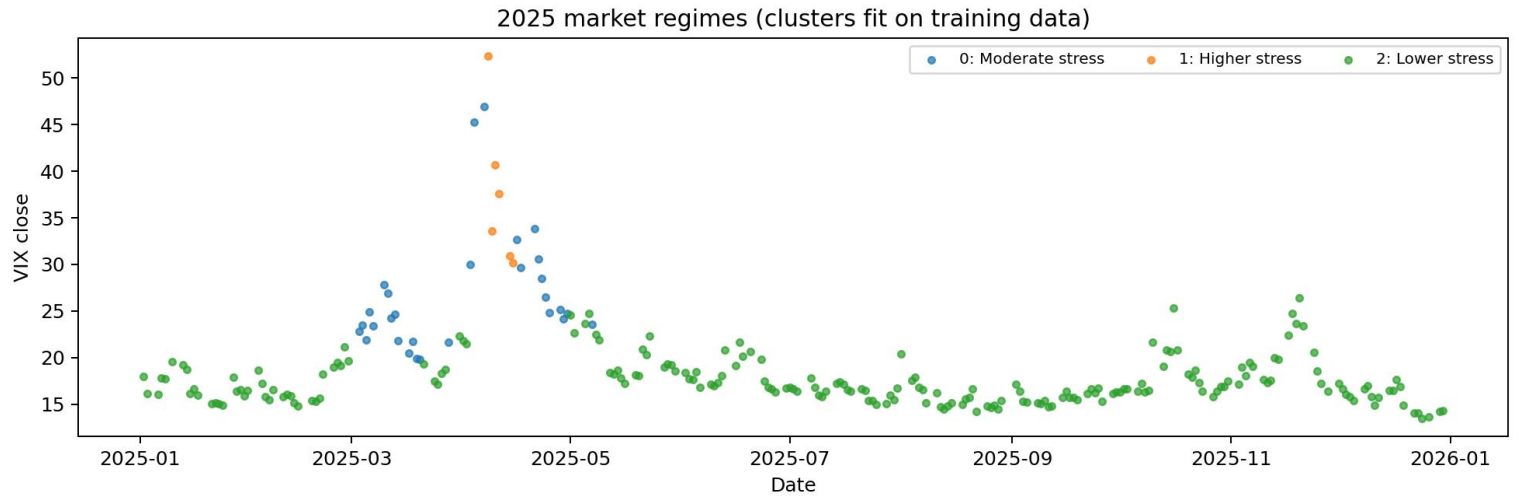
VIX index levels and changes

Market-wide forward volatility expectations

Joined by trading date

**Train: 2018–2024 → Out-of-time test: 2025 | chronological validation**

# Three interpretable market-stress regimes




**6.8%**

Lower-stress frequency

**27.3%**

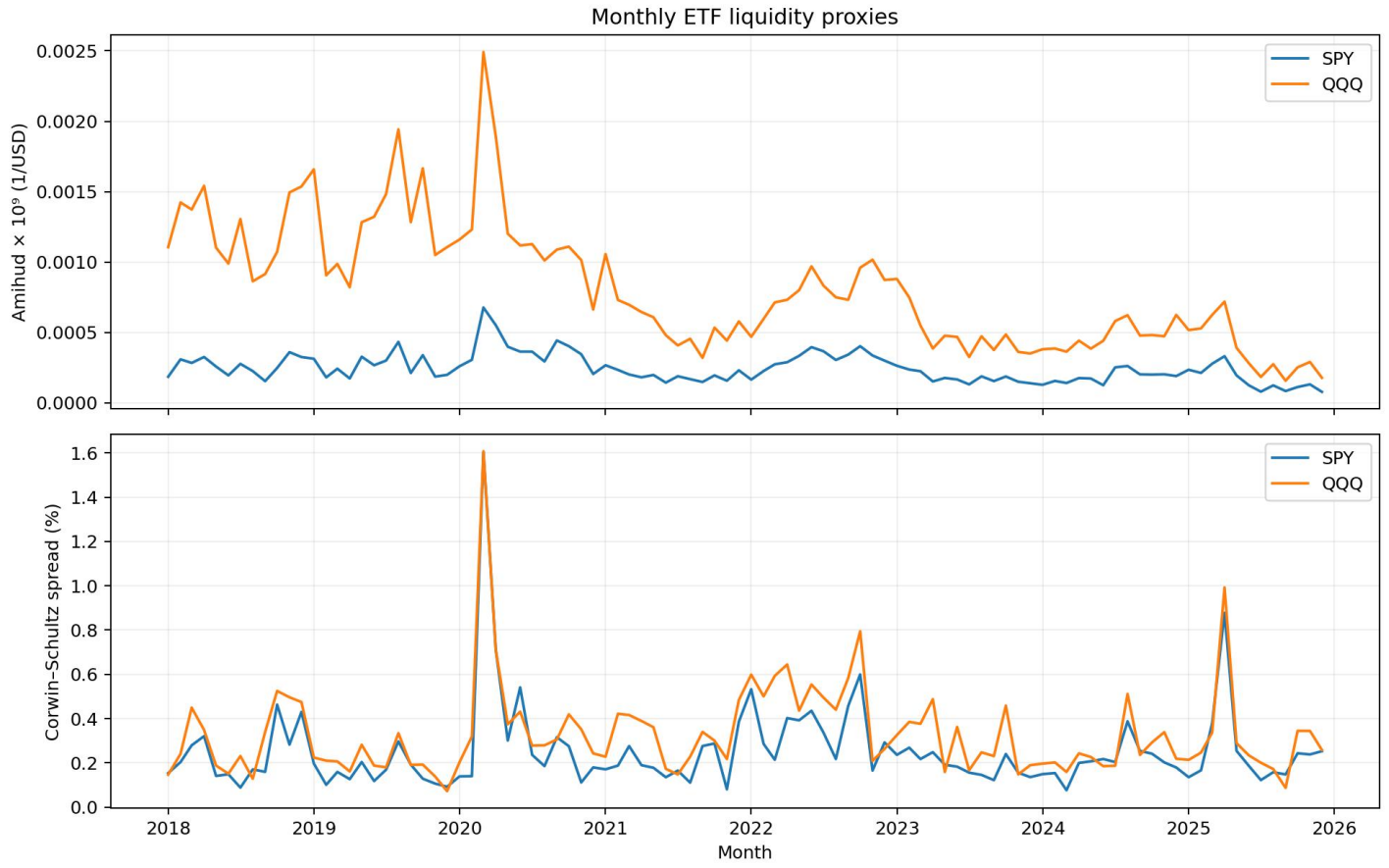
Moderate-stress frequency

**72.0%**

Higher-stress frequency

Training cluster proportions; not calibrated forecasts.

# Trading conditions change across market episodes



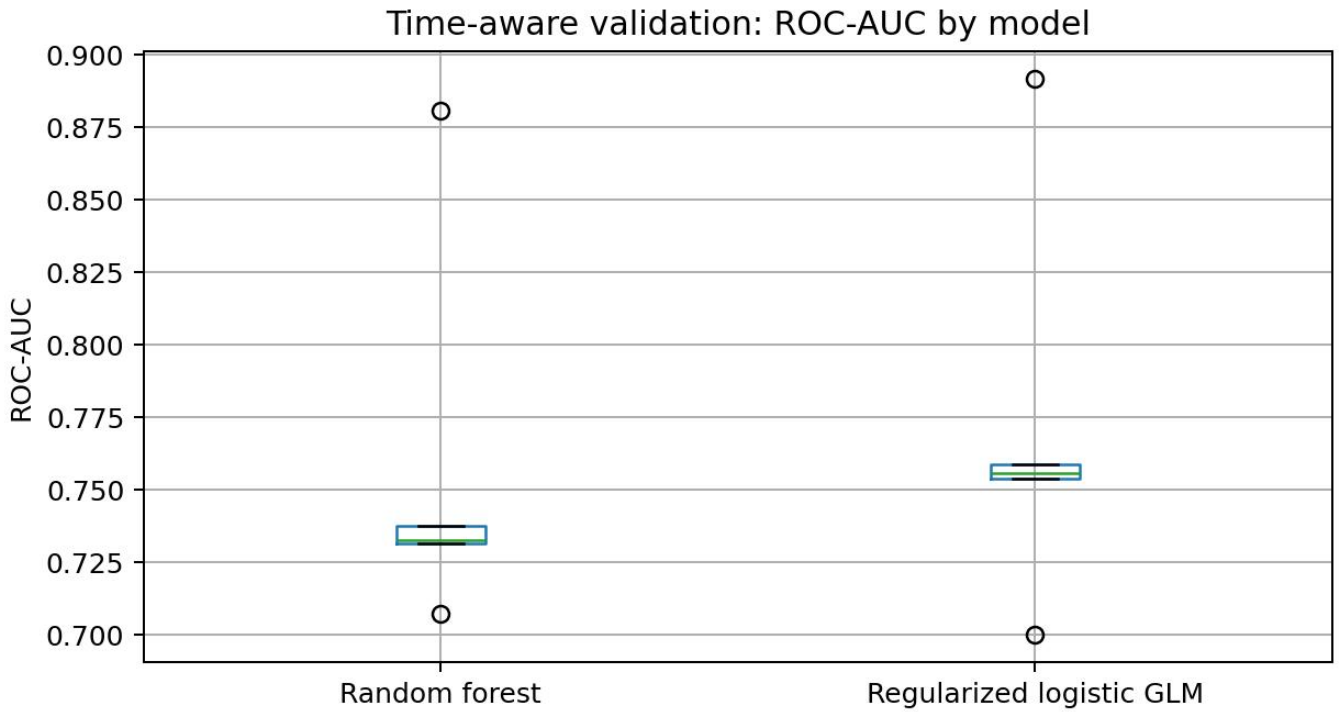

## QQQ

Often higher Amihud illiquidity proxy than SPY.

## 2020

Estimated spreads rose sharply during early-pandemic stress.

# GLM slightly outperforms the ensemble




**0.772**

Logistic GLM mean CV AUC

**0.758**

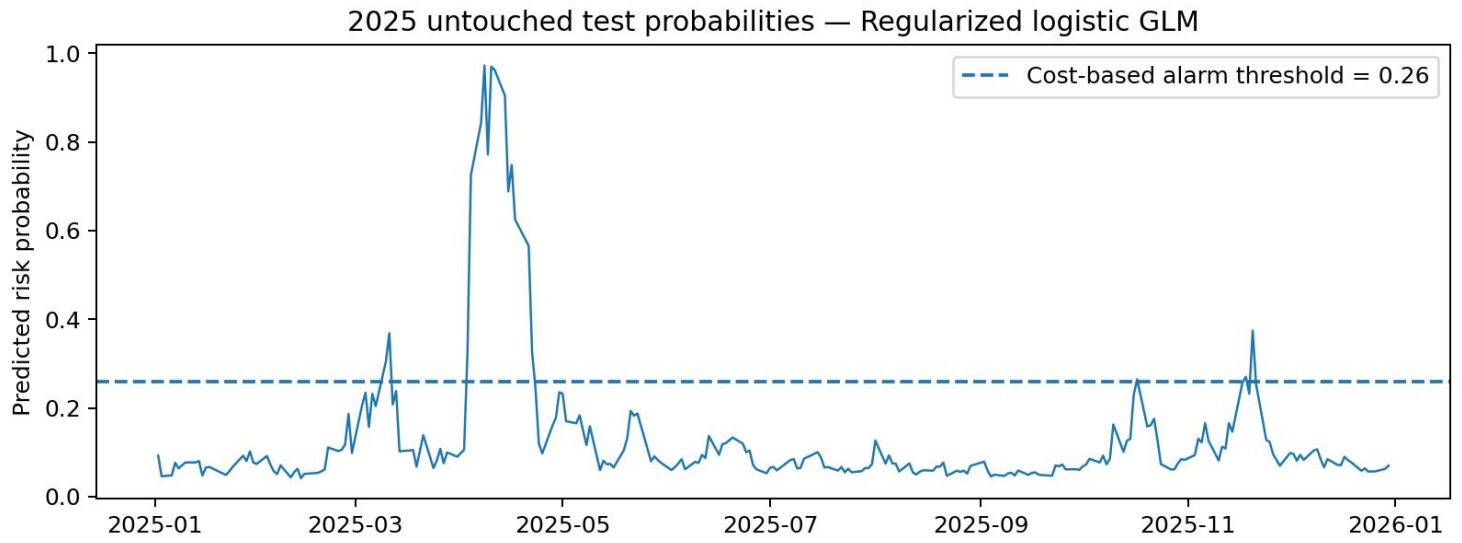
Random forest mean CV AUC

**TUNING CHECK**

GLM C = 0.03

Tuned CV AUC = 0.778

# A good ranking model still misses many risky days




**0.804**

2025 test ROC-AUC

**0.26**

Alert probability threshold

**9 of 33**

Actual high-risk days detected (27% recall)

## Finding 01

Use the GLM as an additional monitoring signal—not an automatic trading rule.

## Finding 02

Liquidity feature removal did not reduce CV AUC: 0.772 → 0.777.

## Limitations

Daily spread estimates are proxies; model recall is low and market regimes can shift.

## Next step

Test new periods, review costs and thresholds, and add richer liquidity data.

Thank you!